Garbage Collection Strategies: Go vs Java vs Julia

Explore how three open source languages—Java, Go, and Julia—take different paths to solve the invisible challenge of garbage collection. Whether you're working with enterprise apps, cloud-native systems, or scientific models, knowing how each language manages memory can help you make better technical decisions.

What happens to your app's memory once it's no longer needed? Most developers don't think about it until performance drops, latency spikes, or crashes creep in.

That's where garbage collection strategies come into play. The way a language handles memory cleanup behind the scenes can quietly shape how fast, reliable, and scalable your software becomes.

Garbage collection isn't about cleaning memory—it's about keeping your software breathing under load.

What is garbage collection?

Garbage collection (GC) is a form of automatic memory management. It's a process that runs in the background of many programming languages, finding and cleaning up memory that's no longer needed so developers don't have to do it manually.

Without garbage collection, unused memory can pile up, leading to memory leaks, slower performance, and system

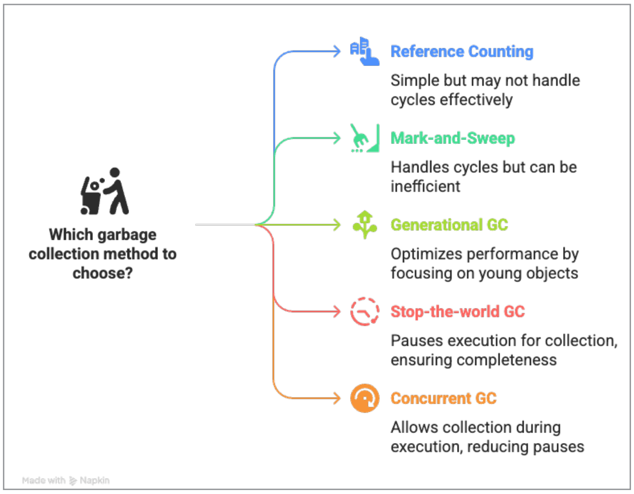

Figure 1: Garbage collection methods